Loading and processing 10x data

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Loading packages

```
suppressMessages(library(dplyr))
suppressMessages(library(Seurat))
suppressMessages(library(patchwork))
suppressMessages(library(ggplot2))
suppressMessages(library(writexl))
suppressMessages(library(SingleR))
suppressMessages(library(scuttle))
suppressMessages(library(scRNAseq))
suppressMessages(library(RColorBrewer))
suppressMessages(library(stringr))
suppressMessages(library(formatR))
```

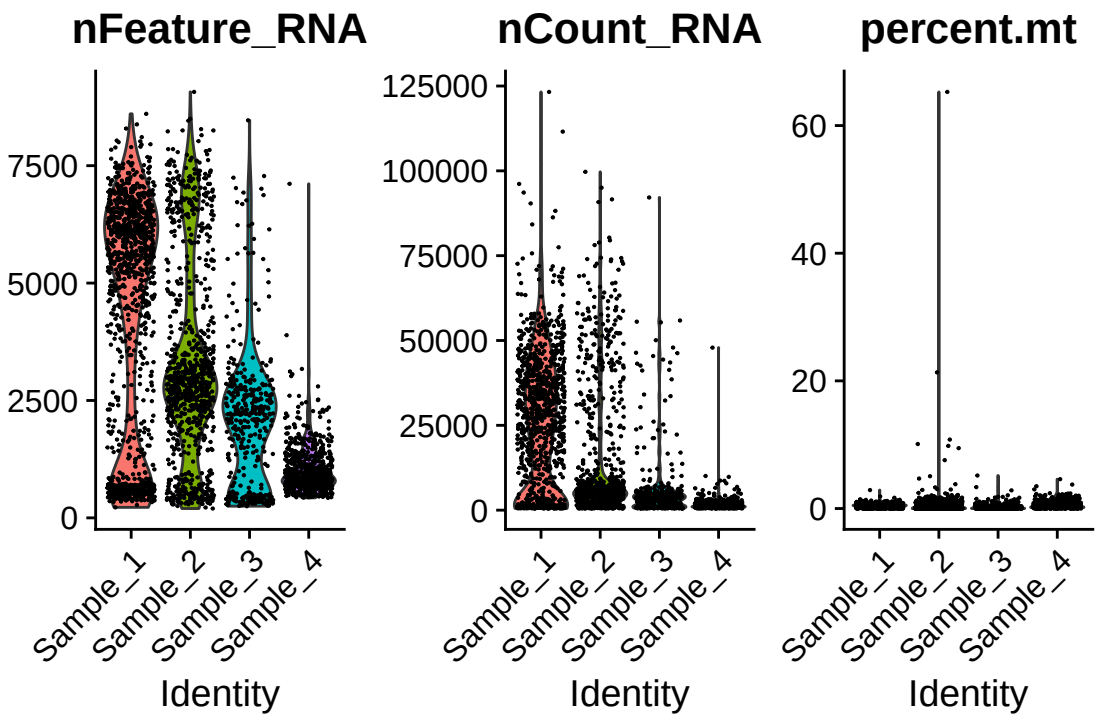
Loading data

```
# The path should be modifie to include your directory of the cellranger output
all_dirs <- dir(path = "Data/outs/per_sample_outs/", full.names = T)

list_sample <- list()
for (i in 1:length(all_dirs)) {

  # here the file path describe the filtered matrix from cellranger output
  Seq_raw_file <- Read10X(data.dir =
paste0(all_dirs[i],"/count/sample_filtered_feature_bc_matrix/"))
  Seurat_file <- CreateSeuratObject(counts = Seq_raw_file, project =
str_sub(all_dirs[i],-8,-1), min.cells = 3, min.features =   150)
  list_sample <- append(list_sample, Seurat_file)
```

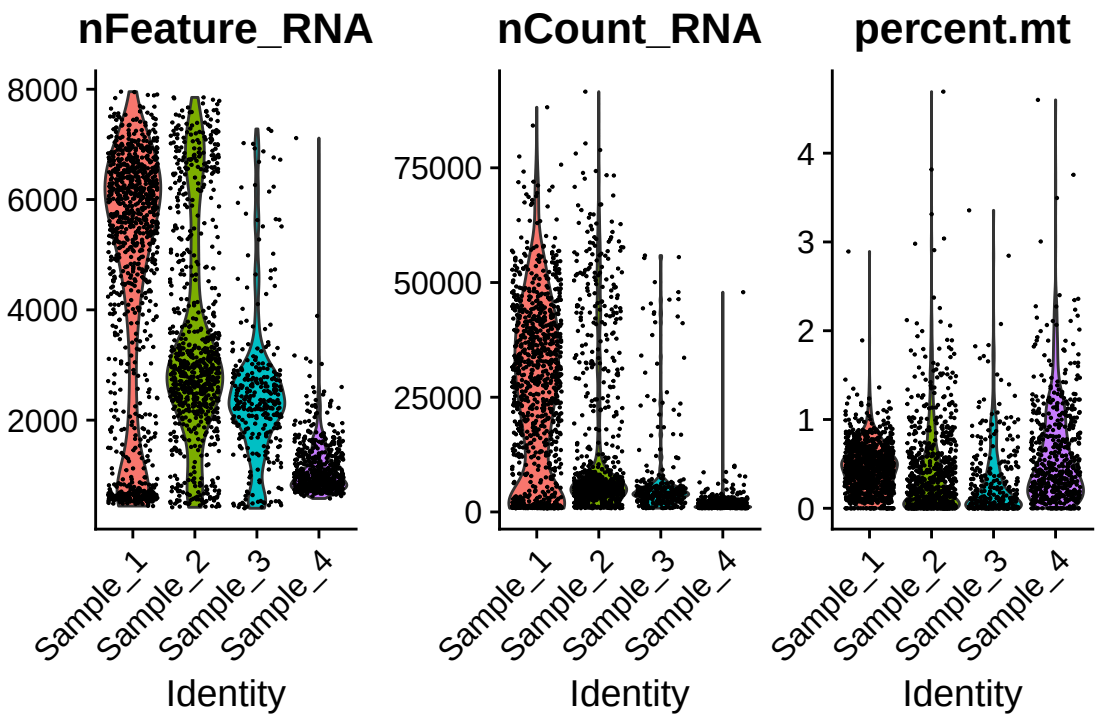
```
}

lavage_cells10x <- merge(list_sample[[1]], y = list_sample[-1], add.cell.ids =
c("1","2","3", "4"), project = "bronchoalveolar_cells")
```

QC

```
lavage_cells10x[["percent.mt"]] <- PercentageFeatureSet(lavage_cells10x, pattern =
"^MT-")
VlnPlot(lavage_cells10x, features = c("nFeature_RNA", "nCount_RNA", "percent.mt"), ncol =
3, pt.size = 0.1)
```




```
lavage_cells10x <- subset(lavage_cells10x, subset = nFeature_RNA < 8000 & nFeature_RNA >
400 & nCount_RNA > 800 & percent.mt < 5)
VlnPlot(lavage_cells10x, features = c("nFeature_RNA", "nCount_RNA", "percent.mt"),
group.by = "orig.ident", ncol = 3, pt.size = 0.1)
```




```
#ggsave("VlnPlot_QC.png", height = 8, width = 12)
```

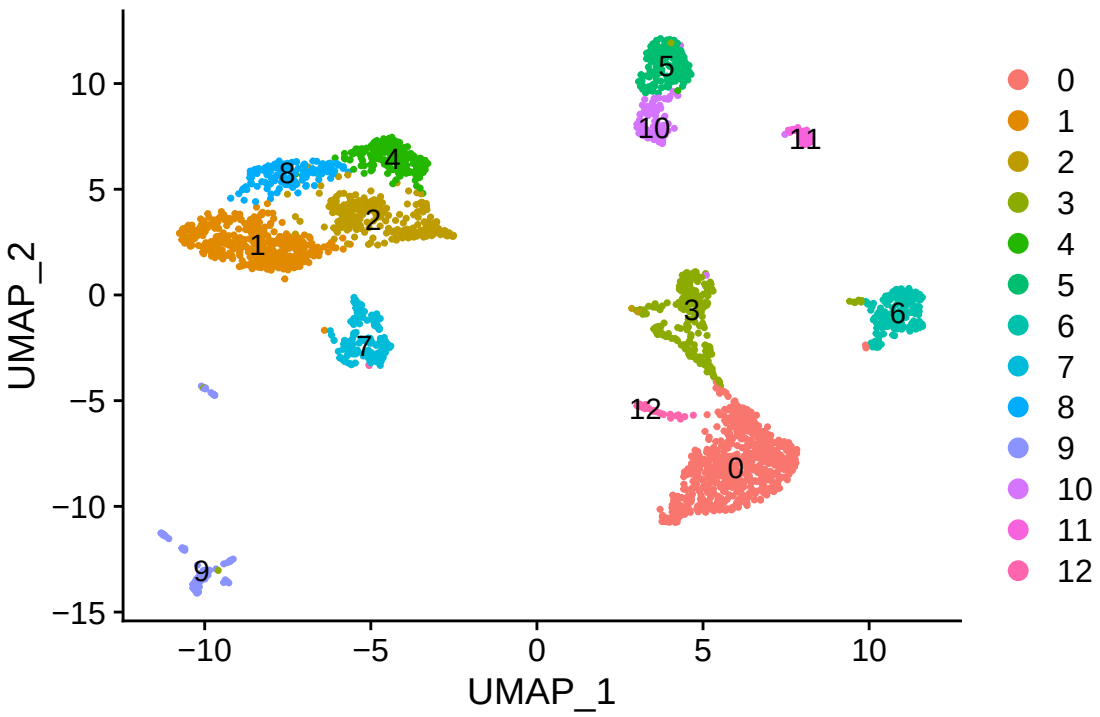
Normalisation + scalin

```
lavage_cells10x <- NormalizeData(lavage_cells10x, normalization.method = "LogNormalize",
scale.factor = 10000)
lavage_cells10x <- FindVariableFeatures(lavage_cells10x, selection.method = "vst",
nfeatures = 2000)

all.genes <- rownames(lavage_cells10x)
lavage_cells10x <- ScaleData(lavage_cells10x, features = all.genes)
```

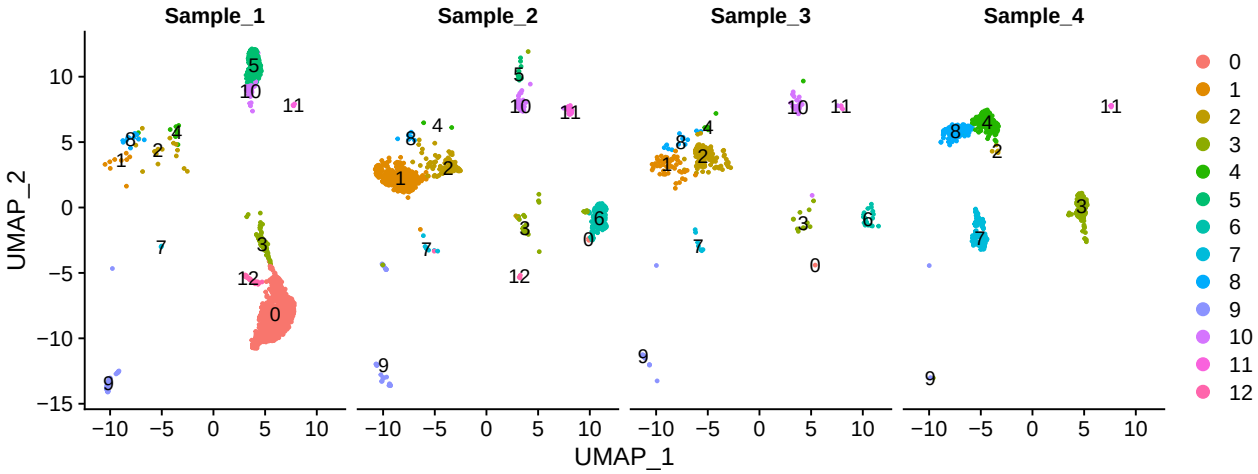
Dimension reduction

```
lavage_cells10x <- RunPCA(lavage_cells10x, features = VariableFeatures(object =
lavage_cells10x))
#DimPlot(lavage_cells10x, reduction = "pca")

lavage_cells10x <- FindNeighbors(lavage_cells10x, dims = 1:16)
lavage_cells10x <- FindClusters(lavage_cells10x, resolution = 0.5)
lavage_cells10x <- RunUMAP(lavage_cells10x, dims = 1:16)
```

```
DimPlot(lavage_cells10x, reduction = "umap", label = T)
```




```
DimPlot(lavage_cells10x, reduction = "umap", split.by = "orig.ident", label = T)
```




```
saveRDS(lavage_cells10x, "lavage_cells10x.rds")
```

```
sessionInfo()
```

```
## R version 4.3.2 (2023-10-31)
## Platform: x86_64-pc-linux-gnu (64-bit)
## Running under: Ubuntu 22.04.3 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
## LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.20.so;  LAPACK version 3.10.0
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
```

```
##  [3] LC_TIME=fr_BE.UTF-8        LC_COLLATE=en_US.UTF-8
##  [5] LC_MONETARY=fr_BE.UTF-8    LC_MESSAGES=en_US.UTF-8
##  [7] LC_PAPER=fr_BE.UTF-8       LC_NAME=C
##  [9] LC_ADDRESS=C               LC_TELEPHONE=C
## [11] LC_MEASUREMENT=fr_BE.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Europe/Brussels
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats4    stats     graphics  grDevices utils     datasets  methods
## [8] base
##
## other attached packages:
##  [1] formatR_1.14                stringr_1.5.0
##  [3] RColorBrewer_1.1-3          scRNAseq_2.14.0
##  [5] scuttle_1.10.1              SingleCellExperiment_1.22.0
##  [7] SingleR_2.2.0               SummarizedExperiment_1.30.2
##  [9] Biobase_2.60.0              GenomicRanges_1.52.0
## [11] GenomeInfoDb_1.36.0         IRanges_2.34.0
## [13] S4Vectors_0.38.1            BiocGenerics_0.46.0
## [15] MatrixGenerics_1.12.2       matrixStats_1.0.0
## [17] writexl_1.4.2               ggplot2_3.4.2
## [19] patchwork_1.1.2             SeuratObject_4.1.3
## [21] Seurat_4.3.0                dplyr_1.1.2
##
## loaded via a namespace (and not attached):
##   [1] RcppAnnoy_0.0.21              splines_4.3.2
##   [3] later_1.3.1                   BiocIO_1.10.0
##   [5] filelock_1.0.2                bitops_1.0-7
##   [7] R.oo_1.25.0                   tibble_3.2.1
##   [9] polyclip_1.10-4               XML_3.99-0.14
##  [11] lifecycle_1.0.3               ensembldb_2.24.0
##  [13] globals_0.16.2                lattice_0.22-5
##  [15] MASS_7.3-60                   magrittr_2.0.3
##  [17] plotly_4.10.2                 rmarkdown_2.23
##  [19] yaml_2.3.7                    httpuv_1.6.11
##  [21] sctransform_0.3.5             spam_2.9-1
##  [23] sp_2.0-0                      spatstat.sparse_3.0-2
##  [25] reticulate_1.30               cowplot_1.1.1
##  [27] pbapply_1.7-2                 DBI_1.1.3
##  [29] abind_1.4-5                   zlibbioc_1.46.0
##  [31] Rtsne_0.16                    R.utils_2.12.2
##  [33] purrr_1.0.1                   AnnotationFilter_1.24.0
##  [35] RCurl_1.98-1.12               rappdirs_0.3.3
##  [37] GenomeInfoDbData_1.2.10       ggrepel_0.9.3
##  [39] irlba_2.3.5.1                 listenv_0.9.0
##  [41] spatstat.utils_3.0-3          goftest_1.2-3
##  [43] spatstat.random_3.1-5         fitdistrplus_1.1-11
##  [45] parallelly_1.36.0             DelayedMatrixStats_1.22.1
##  [47] leiden_0.4.3                  codetools_0.2-19
##  [49] DelayedArray_0.26.3           xml2_1.3.4
##  [51] tidyselect_1.2.0              farver_2.1.1
##  [53] ScaledMatrix_1.8.1            BiocFileCache_2.8.0
```

```
##  [55] spatstat.explore_3.2-1        GenomicAlignments_1.36.0
##  [57] jsonlite_1.8.7                ellipsis_0.3.2
##  [59] progressr_0.13.0              ggridges_0.5.4
##  [61] survival_3.5-7                progress_1.2.2
##  [63] tools_4.3.2                   ica_1.0-3
##  [65] Rcpp_1.0.11                   glue_1.6.2
##  [67] gridExtra_2.3                 xfun_0.39
##  [69] withr_2.5.0                   BiocManager_1.30.21
##  [71] fastmap_1.1.1                 fansi_1.0.4
##  [73] digest_0.6.33                 rsvd_1.0.5
##  [75] R6_2.5.1                      mime_0.12
##  [77] colorspace_2.1-0              scattermore_1.2
##  [79] tensor_1.5                    biomaRt_2.56.1
##  [81] spatstat.data_3.0-1           RSQLite_2.3.1
##  [83] R.methodsS3_1.8.2             utf8_1.2.3
##  [85] tidyr_1.3.0                   generics_0.1.3
##  [87] data.table_1.14.8             rtracklayer_1.60.0
##  [89] prettyunits_1.1.1             httr_1.4.6
##  [91] htmlwidgets_1.6.2             S4Arrays_1.0.4
##  [93] uwot_0.1.16                   pkgconfig_2.0.3
##  [95] gtable_0.3.3                  blob_1.2.4
##  [97] lmtest_0.9-40                 XVector_0.40.0
##  [99] htmltools_0.5.5               dotCall64_1.0-2
## [101] ProtGenerics_1.32.0           scales_1.2.1
## [103] png_0.1-8                     knitr_1.43
## [105] rstudioapi_0.14               rjson_0.2.21
## [107] reshape2_1.4.4                curl_5.0.1
## [109] nlme_3.1-163                  zoo_1.8-12
## [111] cachem_1.0.8                  BiocVersion_3.17.1
## [113] KernSmooth_2.23-22            parallel_4.3.2
## [115] miniUI_0.1.1.1                AnnotationDbi_1.62.1
## [117] restfulr_0.0.15               pillar_1.9.0
## [119] grid_4.3.2                    vctrs_0.6.3
## [121] RANN_2.6.1                    promises_1.2.0.1
## [123] BiocSingular_1.16.0           dbplyr_2.3.2
## [125] beachmat_2.16.0               xtable_1.8-4
## [127] cluster_2.1.5                 evaluate_0.21
## [129] GenomicFeatures_1.52.0        Rsamtools_2.16.0
## [131] cli_3.6.1                     compiler_4.3.2
## [133] rlang_1.1.1                   crayon_1.5.2
## [135] future.apply_1.11.0           labeling_0.4.2
## [137] plyr_1.8.8                    stringi_1.7.12
## [139] viridisLite_0.4.2             deldir_1.0-9
## [141] BiocParallel_1.34.2           munsell_0.5.0
## [143] Biostrings_2.68.1             lazyeval_0.2.2
## [145] spatstat.geom_3.2-4           Matrix_1.6-1
## [147] ExperimentHub_2.8.0           hms_1.1.3
## [149] sparseMatrixStats_1.12.0      bit64_4.0.5
## [151] future_1.33.0                 KEGGREST_1.40.0
## [153] shiny_1.7.4.1                 highr_0.10
## [155] interactiveDisplayBase_1.38.0 AnnotationHub_3.8.0
## [157] ROCR_1.0-11                   igraph_1.5.0.1
## [159] memoise_2.0.1                 bit_4.0.5
```